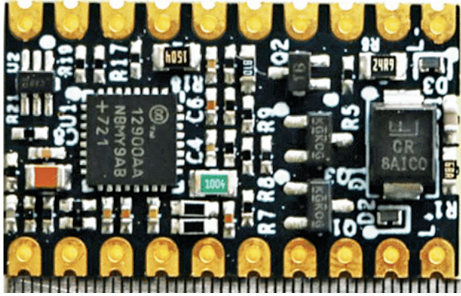
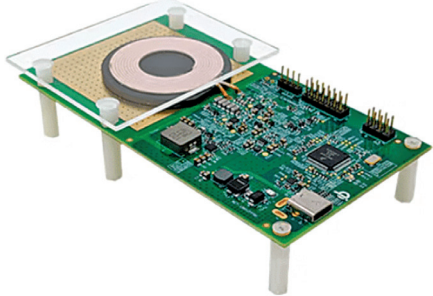


Interesting Reference Designs Of TRANSMITTERS

Title of Reference Design	Current Loop Transmitter	Wireless Power Transmitter
Image		
Key Attractions	The reference design enables you to accurately control and measure current using just a microcontroller, eliminating the need for additional parts. It is simple, reliable, and makes building analogue output systems easier.	The 22W wireless power transmitter reference design lets engineers make safe and standard chargers. It works with Qi 1.3.2, finds metal objects, and uses NFC for secure charging in small devices.
Highlights	The MAXREFDES1161 reference design from Analog Devices (ADI) uses two PWM signals from a microcontroller, one for coarse 8-bit control and the other for fine 8-bit control, to set the current loop output without a DAC. A built-in amplifier measures the loop current and sends a signal to the microcontroller's ADC. The system is powered by a 24V loop supply, with inrush current limited by a resistor and a regulator that steps the voltage down to 3.3V, set by two external resistors. The setup allows analogue output using microcontroller I/O, with filtering handled by the multi-feedback system. Current measurement, a 2.5V reference, and comparator-based checks allow monitoring without extra circuitry. The startup sequence ensures stability before powering external components. The design supports the development of analogue output modules with fewer components and basic microcontroller integration.	The MP-A11 fixed frequency wireless power transmitter reference design from NXP Semiconductors is a 22W solution for mid-power wireless charging in automotive consoles, industrial systems, smart home products, and portable consumer electronics. Built around NXP's MWCT1013A controller, the design supports Qi 1.3.2 compliance and includes full-bridge control, foreign object detection (FOD), and real-time receiver communication. Operating at a fixed frequency of 127.7kHz with an MP-A11 coil, it maintains a small footprint and low electromagnetic interference (EMI) for use in space-constrained designs. The design incorporates FOD, near-field communication (NFC)-based Qi authentication, and a power stage featuring N-channel metal-oxide-semiconductor field-effect transistors (MOSFETs) and gate drivers. The firmware supports over-the-air (OTA) updates and diagnostics through universal asynchronous receiver-transmitter (UART) or inter-integrated circuit (I ² C) to reduce time to market.
Applications	The reference design is suitable for industrial or embedded applications.	The reference design is suitable for wireless charging applications.
OEM Brand	Analog Devices (ADI)	NXP Semiconductors
URL	https://www.electronicsforu.com/electronics-projects/current-loop-transmitter-reference-design	https://www.electronicsforu.com/electronics-projects/wireless-power-transmitter-reference-design-2